



Title: Networking Fundamentals, Nmap Scanning, and Automation Scripting

Introduction

In *ethical hacking* or *penetration testing*, the first phase is *information gathering* about a specific target. This phase helps in mapping the system and determining which tools to use. It also assists in identifying misconfigured or open ports and services, which can later be analyzed for potential exploitation.

This report presents a practical exploration of networking fundamentals, network scanning using *Nmap*, and automation using *Python*. The purpose of this task is to understand how systems communicate over a network, identify open ports and services, analyze potential security risks, and automate the scanning process.

Introduction and Installation of Nmap

Nmap ("*Network Mapper*") is an open-source tool for network exploration and security audit. It uses raw IP packets in novel ways to determine hosts, services, operating system, packet filters/ firewalls are in use in a single or multiple hosts.

Installing Nmap

1. In terminal type " **sudo apt-get install nmap -y**".
 2. After the installation is complete use "**nmap -v**" to check if the tool installed correctly.
 3. Then we can use "**man nmap**" to review the documentation on nmap with its basic commands, it's synopsis and different options to use the tool effectively.
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Reconnaissance Using Nmap

I will use **-oN** at the end of most of the nmap commands to document that scan. **-oN** just puts the scan in a .txt or any kind of extension file or readable file that you want. It just makes it easy to access that scan that, so you don't have to scan the host or the network again with that same option.

1. SYN Scan: **nmap -sS <target ip> -oN syn_scan.txt**

- Sends SYN packets
- Receives SYN-ACK packets
- Does not complete the handshake.
- Faster and stealthier

Response from SYN Scan:



```
nmap -sS 192.168.xxx.xxx -oN syn_scan.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-30 19:28 EDT
Nmap scan report for 192.168.xxx.xxx
Host is up (0.00011s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
3306/tcp   open  mysql
MAC Address: 00:00:00:00:00:00 (VMware)
```

2. TCP Connect Scan: `nmap -sT <target> -oN tcp_scan.txt`

- Completes full 3-way handshake.
- Uses OS networking stack.
- Slower and easier to log.
- Bypasses the TCP filtering.

Response from TCP Connect Scan:

```
nmap -sT 192.168.xxx.xxx -oN tcp_scan.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-30 19:28 EDT
Nmap scan report for 192.168.xxx.xxx
Host is up (0.0013s latency).
Not shown: 997 closed tcp ports (conn-refused)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
3306/tcp   open  mysql
MAC Address: 00:00:00:00:00:00 (VMware)
```

3. UDP Scan: `nmap -sU <target> -oN udp_scan.txt`

- UDP is connectionless protocol.
- No handshake in UDP.
- Slower than TCP scans.

Response from UDP Scan:

```
nmap -sU 192.168.xxx.xxx -oN udp_scan.txt
Nmap scan report for 192.168.xxx.xxx
Host is up (0.00037s latency).
All 1000 scanned ports on 192.168.xxx.xxx are in ignored states.
Not shown: 1000 closed udp ports (port-unreach)
MAC Address: 00:00:00:00:00:00 (VMware)
```

4. OS Detection Scan: `nmap -O <target> -oN os_scan.txt`

- Enables OS detection.
- Finds out the OS of the host.



Response from OS Detection Scan:

```
nmap -O 192.168.xxx.xxx -oN os_scan.txt
Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-30 20:08 EDT
Nmap scan report for 192.168.xxx.xxx
Host is up (0.00025s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE
80/tcp    open  http
443/tcp   open  https
3306/tcp  open  mysql
MAC Address: 00:00:00:00:00:00 (VMware)
Device type: general purpose
Running: Linux 2.4.X
OS CPE: cpe:/o:linux:linux_kernel:2.4
OS details: Linux 2.4.18 - 2.4.35 (likely embedded)
Network Distance: 1 hop
```

Analyzing the Scans and Host

Machine Overview

Badstore.net is dedicated to helping you understand how hackers play on Web application vulnerabilities. This Web Application has some common vulnerabilities like SQL Injection, XSS, Command Injection via Parameter Tempering, etc and with the OS detection scan we can see it runs under Linux environment. It's a Local VM target taken from vulnhub site that stores multiple vulnerable machines to attack and test your penetration testing skills.

Nmap Commands I used for this task

1. `nmap -sS <target> -oN syn_scan.txt` - stealth scan or SYN scan
2. `nmap -sT <target> -oN tcp_scan.txt` - TCP full connect scan
3. `nmap -sU <target> -oN udp_scan.txt` - UDP scan
4. `nmap -O <target> -oN os_scan.txt` - OS detection scan
5. `nmap -sV <target> -oN version_scan.txt` - services and version detection scan

Table of discovered Ports and Services

PORTS	STATE	SERVICE	VERSION
80/tcp	open	http	Apache httpd 1.3.28
443/tcp	open	ssl/http	Apache httpd 1.3.28
3306/tcp	open	mysql	MySQL 4.1.7-standard

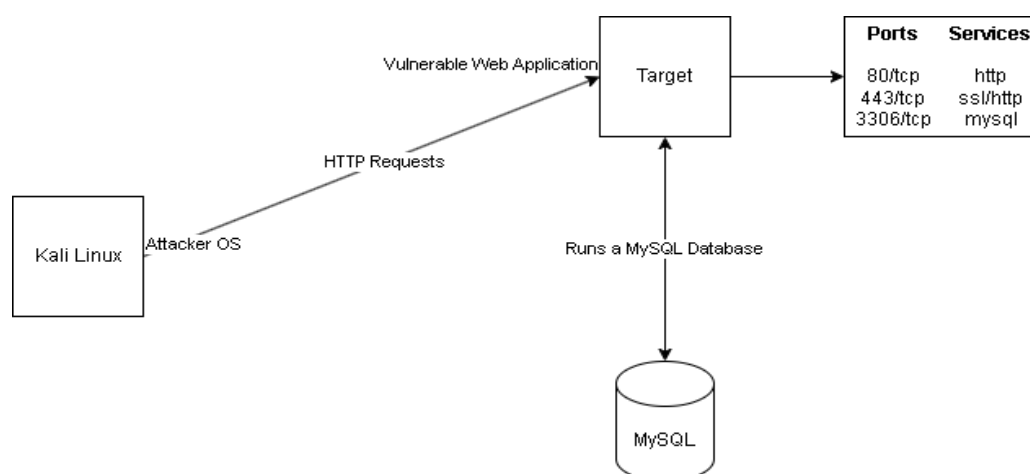
Table 1.1



Potential Security risks and Notes on *http* and *mysql* services

1. HTTP stands for Hyper Text Transfer Protocol. It's a communication protocol used to transfer data between a *client* (like your browser) and a *server* (a website).
 - CVE-2002-0392 which is for Buffer Overflow vulnerability based on the Apache http 1.3.28 which was 2003 version.
 - CVE-2003-1418 crucial vulnerability which can reveal sensitive data via ETag Header which can reveal inode number, etc.
 - Port 80 http is susceptible to XSS which helped me reveal SSoid.
2. MySQL is a database that collects all the user's data that visits the site with their credentials their profile, etc. This database also saves the sites data or metadata in it as well.
 - CVE-2004-0836 allows remote DNS servers to cause a denial of service and possibly execute arbitrary code via a DNS response.
 - CVE-2005-2558 this CVE tells about one of the ways to authentication bypass using stack-based buffer overflow.
 - Port 3306 is susceptible to Authentication Bypass and Privilege Escalation.

Figure 1.1 - Attacker-Target Interaction Diagram



Python Automation Script and Output

```
import nmap from datetime import datetime

def run_scan(target): print(f"[+] Starting scan on {target}...\n")

# Initialize Nmap scanner
scanner = nmap.PortScanner()
```



```
try:
    # Perform SYN scan
    scanner.scan(hosts=target, arguments='-sS -sV')

except Exception as e:
    print(f"[!] Error: {e}")
    return

# Get scan time
scan_time = datetime.now().strftime("%Y-%m-%d %H:%M:%S")

report = []
report.append("=== Nmap Automation Scan Report ===")
report.append(f"Scan Time: {scan_time}")
report.append(f"Target: {target}\n")

# Loop through hosts
for host in scanner.all_hosts():
    report.append(f"Host: {host}")
    report.append(f"State: {scanner[host].state()}")

    for proto in scanner[host].all_protocols():
        report.append(f"\nProtocol: {proto}")

        ports = scanner[host][proto].keys()

        for port in sorted(ports):
            service = scanner[host][proto][port]

            name = service.get('name', 'unknown')
            version = service.get('version', '')
            product = service.get('product', '')

            report.append(
                f"Port: {port} | Service: {name} | Product: {product} |
Version: {version}"
            )

        report.append("\n" + "-"*40 + "\n")

# Save report
with open("scan_report.txt", "w") as f:
    f.write("\n".join(report))
```



```
print("[+] Scan complete! Report saved as scan_report.txt")
```

```
if name == "main": target = input("Enter target IP or hostname: ") run_scan(target)
```

Output of the Script Scan

```
cat scan_report.txt
```

```
=== Nmap Automation Scan Report ===
```

```
Scan Time: 2026-03-31 06:18:28
```

```
Target: 192.168.xxx.xxx
```

```
Host: 192.168.xxx.xxx
```

```
State: up
```

```
Protocol: tcp
```

```
Port: 80 | Service: http | Product: Apache httpd | Version: 1.3.28
```

```
Port: 443 | Service: http | Product: Apache httpd | Version: 1.3.28
```

```
Port: 3306 | Service: mysql | Product: MySQL | Version: 4.1.7-standard
```

Conclusion

The task successfully demonstrated how network reconnaissance and scanning are essential first steps in ethical hacking. Using Nmap, open ports 80, 443, and 3306 and their associated services were identified, along with the target system's operating system (Linux).

Key Learnings

- Learned to do SYN, TCP full connect and UDP scan using nmap.
- Learned to make a data flow diagram using app.diagrams.net.
- Get to know about services common vulnerabilities by a CVE MITRE huge database.
- Learned about XSS and SQL Injection and performed a basic XSS in the target.